National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Hepatitis C: RNA (HCV-RNA), Confirmed Antibody (INNO-LIA), & Genotype (HEPC_L)

Data File: HEPC_L.xpt

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Component Description

Hepatitis viruses constitute a major public health problem because of the morbidity and mortality associated with the acute and chronic consequences of these infections. Because of the high rate of asymptomatic infection with these viruses, information about the prevalence of these diseases is needed to monitor prevention efforts. By testing a nationally representative sample of the U.S. population, NHANES provides the most reliable estimates of age-specific prevalence needed to evaluate the effectiveness of the strategies to prevent these infections. NHANES testing for markers of infection with hepatitis viruses is used to determine secular trends in infection rates across most age and racial/ethnic groups, and provides a national picture of the epidemiologic determinants of these infections.

Hepatitis is inflammation of the liver most often caused by a virus. Viral hepatitis is a major public health problem of global importance because of the ongoing transmission of viruses that cause the disease and increased morbidity and mortality associated with the acute and chronic consequences of these infections. Recommendations have also been developed for the prevention and control of hepatitis C virus (HCV) infection. Global and U.S. goals have been established for elimination of viral hepatitis as a public health threat by 2030 (HHS Healthy People, 2022 and HHS 2020).

In the U.S., the most common types of viral hepatitis are hepatitis A, B, and C. Effective vaccines are available to help prevent hepatitis A and hepatitis B. No vaccine is available for hepatitis C; however, highly effective, well-tolerated treatment can cure hepatitis C virus infection. Hepatitis D virus infection is less common in the U.S. and can occur only among people with hepatitis B virus infection. Hepatitis E infection also is less common in the U.S. These five hepatitis viruses, also called hepatitides, are well-characterized for detection with laboratory assays and are monitored in U.S. public health surveillance systems.

NHANES viral hepatitis data are used to monitor progress toward goals in *Healthy People* and the Health and Human Services (HHS) *Viral Hepatitis National Strategic Plan*, which in turn support U.S. and global viral hepatitis elimination goals (HHS Healthy People, 2022 and National Academies of Sciences, Engineering, and Medicine, 2017). The viral hepatitis laboratory and interview components of NHANES complement data from outbreak, case-based surveillance, vital statistics, health care systems, and cohort studies that can provide timely, detailed, or longitudinal information for subnational geographic areas and disproportionately affected populations, such as people experiencing homelessness or living in correctional facilities; however, these sources lack information available from NHANES, such as race, ethnicity, education, income, and health status and behaviors.

Viral hepatitis data from NHANES are available beginning with NHANES II conducted in 1976-1980 for hepatitis A and hepatitis B, and with NHANES III conducted in 1988-1994 for hepatitis C, hepatitis D, and hepatitis E.

In 2013, CDC revised its guidelines for Hepatitis C (HCV) testing because of 1) changes in the availability of certain commercial HCV antibody tests; 2) evidence that many people who are identified as reactive by an HCV antibody test might not subsequently be evaluated to determine if they have current HCV infection; and 3) there have been significant advances in the development of antiviral agents with improved efficacy against HCV.[1]

The flow chart for the new Hepatitis C algorithm can be found in the laboratory method file or by following the MMWR link found in the references.

Eligible Sample

Examined participants aged 6 years or older were eligible.

Description of Laboratory Methodology

Hepatitis C RNA (HCV-RNA)

The COBAS AmpliPrep COBAS TaqMan Test, is a nucleic acid amplification test for the quantitation of Hepatitis C Virus RNA in human serum or plasma on the COBAS TaqMan 48 or the COBAS 5800 Analyzer.

Hepatitis C Confirmed Antibody (INNO-LIA)

In 2012 and earlier years, the hepatitis C virus testing algorithm began with an antibody screening test, and antibody screening reactive samples were then tested with a recombinant/synthetic immunoblot assay as the antibody confirmation test. Samples with positive antibody confirmation results were then tested with a nucleic acid amplification test for hepatitis C virus RNA. HCV-RNA positive samples were then tested with a line probe assay to determine the virus genotype from six genotypes, and subtype for genotype 1, the most prevalent genotype in the U.S.

Beginning in 2013, samples reactive to the antibody screening test were first tested for RNA. Then, all RNA positive samples were tested for genotype, and only RNA negative samples were tested with an antibody confirmation test. This change in the testing algorithm was made to align the NHANES HCV testing algorithm with a 2013 update to CDC's guidance on testing for HCV infection by clinicians and laboratorians. Additionally, the antibody confirmation test changed to a 3rd generation line immunoassay in 2013 because the test used in 2012 and earlier was no longer commercially available and the stored supply of these test kits had been exhausted.

For the antibody confirmation test, the manual 16-hour sample incubation test procedure is used. 10ul of specimens and controls are diluted in 1mL diluent in test troughs to ensure test strips will slide easily and remain submerged during incubation. Specimens and controls are incubated with agitation by rocker at 34 rpm overnight (16±2 hours) at room temperature (18–25°). After a 3X/5min each wash step on microtiter plate shaker, conjugate is added with a 30-minute incubation at room temperature, followed by aspiration, 3X/5min each wash step and 30-minute incubation in substrate solution also on microtiter plate shaker. Finally, strips are agitated with stop solution for 10–30 minutes at room temperature. The microtiter plate shaker is set for 160 rpm and used for all the washes and solution incubations. Strips are then dried completely in a dry incubator at 37°C for 30 minutes then mounted on the line immunoassay score data reporting sheet with inverted tape for reading within one hour using the reading card. The strips have lines for a background control, three reference lines for antibody (level ±, human IgG, weak positive; level 1+ human IgG, moderate positive; and level 3+, anti-human Ig, strong positive), and six lines for HCV antigens (C1, C2, E2, NS3, NS4, and NS5). Samples with a positive antibody confirmation test result are reported as positive. Samples with a negative antibody confirmation test result are reported as negative. Samples with an indeterminate antibody confirmation test result are released as missing. Only specimens insufficient in quantity are deemed indeterminate.

Hepatitis C genotype

The GenMark Dx eSensor HCVg Direct Test is designed to genotype a panel of eight (8) prevalent HCV types/subtypes (1a, 1b, 2a/c, 3, 4, 5 and 6) in human serum or EDTA plasma samples. Subtype information is available in the majority of cases.

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

There were no changes to laboratory methods, lab equipment, or lab site for this component in the August 2021-August 2023 cycle.

Laboratory Method Files

Hepatitis C Confirmed Antibody Laboratory Procedure Manual (February 2026)

Hepatitis C Genotype Laboratory Procedure Manual (February 2026)

Hepatitis C virus RNA Laboratory Procedure Manual (February 2026)

Laboratory Quality Assurance and Monitoring

Serum samples were processed, stored, and shipped to the Division of Viral Hepatitis, National Center for HIV/AIDS, Viral Hepatitis, STD, and TB Prevention, Centers for Disease Control and Prevention, Atlanta, GA for analysis.

Detailed instructions on specimen collection and processing are discussed in the NHANES Laboratory Procedures Manual (LPM). Vials were stored under appropriate frozen (–30°C) conditions until they were shipped to Division of Viral Hepatitis, National Center for HIV/AIDS, Viral Hepatitis, STD, and TB Prevention for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Act mandates. Detailed QA/QC instructions are discussed in the NHANES LPM.

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the quality-control procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels are maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected on “dry run” sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

Analytic Notes

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. Analysts should evaluate the extent of missing data in the dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES Analytic Guidelines and the on-line NHANES Tutorial for further details on the use of sample weights and other analytic issues.

Phlebotomy Weights

For the August 2021-August 2023 cycle, analysis of nonresponse patterns for the phlebotomy component in the MEC examination revealed differences by age group and race/ethnicity, among other characteristics. For example, approximately 67% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults aged 18 and older provided a blood specimen. Therefore, an additional phlebotomy weight, WTPH2YR, has been included in this data release to address possible nonresponse bias. Participants who are eligible but did not provide a blood specimen have their phlebotomy weight assigned a value of “0” in their records. The phlebotomy weight should be used for analyses that use variables derived from blood analytes, and is included in all relevant data files.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The NHANES August 2021-August 2023 Demographics File contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

This data is qualitative. The use of lower limits of detection (LLODs) is not applicable.

References

- Centers for Disease Control and Prevention. Recommendations for prevention and control of hepatitis C virus (HCV) infection and HCV-related chronic disease. MMWR 1998;47(No. RR-19):1-39. Available from: https://www.cdc.gov/mmwr/PDF/RR/RR4719.pdf.
- National Academies of Sciences, Engineering, and Medicine. 2017. A national strategy for the elimination of hepatitis B and C. Washington, DC: The National Academies Press. Available from: http://www.nationalacademies.org/hmd/reports/2017/national-strategy-for-the-elimination-of-hepatitis-b-and-c.aspx
- Testing for HCV Infection: An Update of Guidance for Clinicians and Laboratorian. Morbidity and Mortality Weekly Report. Centers for Disease Control and Prevention. May 7, 2013. <https://www.cdc.gov/mmwr/pdf/wk/mm62e0507a2.pdf>.
- U.S. Department of Health and Human Services. 2020. Viral Hepatitis National Strategic Plan for the United States: A Roadmap to Elimination (2021–2025). Washington, DC. Available from: https://www.hhs.gov/hepatitis/viral-hepatitis-national-strategic-plan/index.html
- U.S. Department of Health and Human Services. Healthy People. 2022. Available from: https://health.gov/our-work/national-health-initiatives/healthy-people.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 6 YEARS - 150 YEARS

WTPH2YR - Phlebotomy 2 Year Weight

Variable Name: WTPH2YR

SAS Label: Phlebotomy 2 Year Weight

English Text: Phlebotomy 2 Year Weight

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
4391.8220579 to 241728.85724	Range of Values	7316	7316	
0	No blood sample provided	752	8068	
.	Missing	0	8068	

LBXHCR - Hepatitis C RNA

Variable Name: LBXHCR

SAS Label: Hepatitis C RNA

English Text: Hepatitis C RNA

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Positive	27	27	
2	Negative	113	140	
3	Negative Screening HCV Antibody	6975	7115	
.	Missing	953	8068	

LBDHCI - Hepatitis C Antibody (confirmed)

Variable Name: LBDHCI

SAS Label: Hepatitis C Antibody (confirmed)

English Text: Hepatitis C Antibody (confirmed)

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Positive	87	87	
2	Negative	17	104	
3	Negative Screening HCV Antibody	6975	7079	
4	Positive HCV RNA	27	7106	
.	Missing	962	8068	

LBXHCG - Hepatitis C Genotype

Variable Name: LBXHCG

SAS Label: Hepatitis C Genotype

English Text: Hepatitis C Genotype

Target: Both males and females 6 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Genotype 1a	10	10	
2	Genotype 1b	3	13	
3	Gen 1 other than a/b/not determined	1	14	
4	Genotype 2	0	14	
5	Genotype 3	4	18	
6	Genotype 4	0	18	
7	Genotype 5	0	18	
8	Genotype 6	0	18	
9	Genotype undetermined	0	18	
.	Missing	8050	8068	